

1 SILO Supplementary Material
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4 This supplementary package provides detailed results supporting the paper:

6 "SILO: A Spectre-in-the-Loop CAD Framework for Variability-Aware RFIC Design"

8 The material includes extended performance data, statistical analyses, and reproducibility information for both case studies presented in the paper: a 5 GHz LC oscillator and a 920 MHz RF-DC rectifier.

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12 Structure Overview
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- 15 1. SILO_Overall_Performance_Summary.pdf
16 Provides a consolidated comparison of DE, GA, and Hybrid optimization results across both circuits.
- 18 2. LC_Oscillator/
19 Contains detailed results for the LC oscillator case study.
- 21 3. RF_Rectifier/
22 Contains detailed results for the RF-DC rectifier case study.

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26 Directory Structure
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29 The organization of the supplementary material is as follows:

31 SILO_Supplementary.zip

32 |
33 |--- SILO_Overall_Performance_Summary.pdf
34 |--- README.pdf

36 |--- LC_Oscillator/

37 | |--- DE/

38 | |--- DE_Report.pdf
39 | |--- setup.txt
40 | |--- raw_data/

42 | |--- GA/

43 | |--- GA_Report.pdf
44 | |--- setup.txt
45 | |--- raw_data/

47 | |--- Hybrid/

48 | |--- Hybrid_Report.pdf
49 | |--- setup.txt
50 | |--- raw_data/

52 |--- RF_Rectifier/

53 | |--- DE/

54 | |--- DE_Report.pdf
55 | |--- setup.txt
56 | |--- raw_data/

58 | |--- GA/

59 | |--- GA_Report.pdf
60 | |--- setup.txt
61 | |--- raw_data/

63 | |--- Hybrid/

64 | |--- Hybrid_Report.pdf
65 | |--- setup.txt
66 | |--- raw_data/

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68 Each optimizer folder contains a summary report, simulation setup, and raw data for
reproducibility.
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72 Contents Description
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75 Each optimizer folder (DE, GA, Hybrid) includes:
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77 • Report.pdf
78 A summarized document containing:
79 - Performance metrics (mean and standard deviation)
80 - Monte Carlo (MC) analysis
81 - Process-Voltage-Temperature (PVT) analysis
82 - Key observations and trends
83
84 • setup.txt
85 Simulation configuration and testbench conditions used in Cadence Spectre.
86
87 • raw_data/
88 Detailed datasets for reproducibility, including parameter sweeps, transient data, and
statistical results.
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92 Simulation Details
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95 All results are obtained using transistor-level Cadence Spectre simulations in a 180-nm
CMOS technology.
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97 - Supply Voltage: 1.8 V
98 - PVT Corners: TT, SS, FF, SF, FS
99 - Voltage Range: 1.62 V - 1.98 V
100 - Temperature Range: -40°C to 85°C
101 - Monte Carlo Samples: 1000
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103 The same simulation environment, device models, and testbench configurations are used
for both optimization and verification to ensure signoff consistency.
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107 Notes
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110 - The supplementary material is intended to provide transparency and reproducibility of
the SILO framework results.
111 - Raw datasets are included for completeness; however, summarized results in the PDF
reports should be used for primary interpretation.
112 - All data corresponds directly to the results presented in the main manuscript.
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116 Contact
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119 For any questions regarding the supplementary material, please contact:
120
121 Ahmed Reda Mohamed
122 Interdisciplinary Research Center for Communication Systems and Sensing
123 King Fahd University of Petroleum and Minerals (KFUPM)
124 Email: ahmed.mohamed@kfupm.edu.sa
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